Movie Search App – Project Report

Project Title: Faheem's Reel Explorer

Technology Stack: React (Create React App), JavaScript, CSS, OMDb API, Netlify

Deployment

GitHub Repository

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1. Introduction

Faheem's Reel Explorer is a React-based Movie Search Web Application designed to help users search, explore, and view detailed information about movies in a simple and interactive way. The project focuses on strengthening frontend development skills, understanding component-based architecture, and working with real-world APIs.

2. Purpose of Building This Project

This project was built to:

- Practice React fundamentals such as components, props, and state management.

- Learn how to fetch and display real-time data using a public API.

- Understand UI/UX principles for responsive and interactive web apps.

- Gain hands-on experience in deploying a live project using Netlify.

- Build a portfolio-ready project that demonstrates practical frontend skills.

3. Features and Functionalities

- Search movies by name using a live API.

- Display movie results as interactive cards.

- Show movie posters, titles, and release years.

- Open a modal with detailed movie information on click.

- Responsive design for desktop and mobile devices.

- Clean and modular React component structure.

4. Real-World Applications and Problems Solved

This application demonstrates how real-world search systems work, similar to platforms like IMDb, Netflix search, or OTT discovery tools. It solves problems such as:

- Quick access to movie information.
- Reducing manual searching across multiple platforms.
- Providing a user-friendly interface for content discovery.
- Demonstrating how APIs power modern web applications.

5. Project Structure Overview

The project follows a clean and scalable React structure:

- App.js: Handles search logic, API calls, and state management.
- MovieCard.js: Displays individual movie cards.
- MovieModal.js: Shows detailed movie information.
- index.js: Bootstraps the React application.
- CSS files: Handle styling and responsiveness.

6. Challenges Faced During Development

Some challenges faced while building this project include:

- Understanding asynchronous API calls and handling responses.
- Managing React state and passing data between components.
- Handling UI issues such as modal sizing and responsiveness.
- Debugging API errors and empty search results.
- Deployment configuration and build settings on Netlify.

7. Deployment and Hosting

The project was deployed using Netlify, connected directly to the GitHub repository. Netlify handles build automation, hosting, and continuous deployment, making it easy to update the app with every code push.

8. Future Enhancements and Evolution

This project can be evolved into a full-scale application by adding:

- User authentication and watchlists.
- Pagination and infinite scrolling.
- Advanced filtering by genre, year, or rating.
- Performance optimizations and debounced search.

- Progressive Web App (PWA) features.
- Integration with multiple movie or OTT APIs.

9. Conclusion

Faheem's Reel Explorer is more than a basic search app. It demonstrates real-world frontend development concepts, API integration, and deployment practices. The project lays a strong foundation for building scalable, production-ready web applications in the future.